SAE-Guided Modality-Specific Unlearning in Vision-Language Models

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Abstract

We investigate whether Sparse Autoencoders (SAEs) can enable *surgical, weight-free* concept unlearning inside a Vision-Language Model (VLM). Concretely, we target the *zebra* concept in LLaVA-NeXT-8B and compare three intervention strategies—zero ablation, negative clamping, and additive decoder-weight steering—applied independently to the language model (LM) and the CLIP vision encoder. On the language side, additive SAE steering at Layer 24 with the top-20 zebra features and suppression coefficient $\alpha = -4.4$ achieves an **85.7-point** reduction in Forget Accuracy (FA: $98.7\% \rightarrow 13.0\%$), outperforming Language Gradient Ascent (35.4-point reduction) while maintaining Retain Accuracy at 50.0%. On the vision side, our custom L1-ReLU SAE trained on CLIP Layer 18 achieves complete zebra suppression (FA $= 0\%$) with as few as 20 features at $\alpha = -2.0$. We characterise a *Goldilocks zone* for the suppression coefficient, document cross-modal knowledge leakage, and provide geometric evidence of feature superposition as the mechanism underlying collateral damage to related concepts (e.g. crosswalk recognition).

1. Introduction

Machine unlearning—the removal of specific knowledge from a trained model without full retraining—has become a pressing concern for large generative models, where factual errors, copyrighted content, or harmful concepts may need to be excised post-deployment. Gradient-based approaches such as Gradient Ascent (GA) and Negative Preference Optimisation (NPO) modify model weights globally, often degrading unrelated capabilities and leaving no interpretable audit trail of what was changed.

Sparse Autoencoders (SAEs) (**??**) offer an alternative: they decompose dense residual-stream activations into sparse, near-monosemantic features, each corresponding approximately to one human-interpretable concept. This structure makes it possible to *surgically* suppress a target concept by steering the relevant decoder directions—without touching model weights, and with full reversibility.

SAEs were initially developed for language models. More recently, they have been extended to vision encoders: the ExplainableML/sae-for-vlm work (**?**) trained SAEs on CLIP ViT-L/14-336 and showed that intervening on visual features directly changes the text output of LLaVA. Matryoshka SAEs (**?**) and multimodal SAEs (**?**) expanded coverage further.

Our contribution. We ask: *can SAE-based steering achieve meaningful concept unlearning in a VLM, and what happens when only one modality is steered?* Specifically, we:

- Compare three intervention strategies—zero ablation, negative clamping, and additive steering—for LM-side unlearning, showing additive steering decisively outperforms the others and Gradient Ascent.
- Train custom L1-ReLU SAEs on CLIP ViT-L/14-336, identify Layer 18 as the optimal intervention depth after a systematic layer sweep, and evaluate vision-side unlearning.
- Characterise the *Goldilocks zone* for the suppression coefficient α, beyond which token collapse occurs.
- Provide geometric evidence of *feature superposition* to explain why crosswalk recognition is partially damaged by zebra suppression.
- Measure cross-modal knowledge leakage: after language-only SAE steering, Language Leakage drops

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from 96.0% to 60.0% while Retain Accuracy is preserved at 50.0%.

2. Related Work

Machine Unlearning. TOFU (**?**) and MUSE (**?**) established benchmarks for evaluating forget quality, retain accuracy, and membership-inference resistance in LLMs. MultiDelete (**?**) extended gradient-based unlearning to the multimodal setting.

SAE-Based Steering. **?** showed that SAE steering outperforms zero ablation for concept removal in language models, motivating our use of additive steering. **?** applied CLIP SAE steering to defend against typographic attacks.

Vision SAEs. **?** (NeurIPS 2025) trained L1-ReLU SAEs on CLIP ViT-L/14 and showed that intervening on visual SAE features changes LLaVA's text output—the exact property we exploit. Concurrent with our work, SAUCE applied a similar pipeline at the *penultimate* CLIP layer (Layer 22). We independently found Layer 22 to be suboptimal, and identified Layer 18 as yielding richer, more concept-specific features for unlearning.

3. Background: Sparse Autoencoders

An SAE maps an activation vector $\mathbf{x} \in \mathbb{R}^d$ to a sparse latent code $\mathbf{h} \in \mathbb{R}^n$ ($n \gg d$) and reconstructs $\hat{\mathbf{x}}$:

$$\mathbf{h} = \mathrm{ReLU}(\mathbf{W}_{\mathrm{enc}}(\mathbf{x} - \mathbf{b}_{\mathrm{dec}}) + \mathbf{b}_{\mathrm{enc}}), \tag{1}$$

$$\hat{\mathbf{x}} = \mathbf{W}_{\mathrm{dec}}\,\mathbf{h} + \mathbf{b}_{\mathrm{dec}}. \tag{2}$$

The L1-ReLU SAE is trained with:

$$\mathcal{L} = \underbrace{\|\mathbf{x} - \hat{\mathbf{x}}\|^2}_{\text{reconstruction}} + \lambda\,\|\mathbf{h}\|_1, \tag{3}$$

where λ controls the sparsity–fidelity trade-off. Each column $\mathbf{w}_i$ of $\mathbf{W}_{\mathrm{dec}}$ is a *feature direction*: h_i measures how strongly concept i is present in $\mathbf{x}$.

4. Methodology

4.1. Model Stack

4.2. SAE Training for the Vision Encoder

We trained custom SAEs on frozen CLIP ViT-L/14-336 activations extracted from up to 20,000 ImageNet images with $5\times$ zebra oversampling (class 340), following a three-phase pipeline:

1. **Extract:** Run CLIP forward passes; save 576 patch activations per image per target layer as chunk files.
2. **Stats:** Compute per-layer mean vector and mean L2 norm for normalisation.
3. **Train:** Load normalised chunks and train the SAE.

Table 1. Model and SAE configuration.

Component	Value
VLM backbone	LLaVA-NeXT-8B
Vision encoder	CLIP ViT-L/14-336
Encoder hidden dim	$d = 1024$
Patches per image	576 (CLS dropped)
Vision SAE type	L1-ReLU, $\lambda = 0.03$
Vision SAE layer	Layer 18 (HS index 19)
Vision SAE width	$1024 \rightarrow 32{,}768$
sae_mean_norm	12.11
mean_vector_norm	5.50
Language SAE	EvolvingLMMs-Lab
Language SAE layer	Layer 24
Forget concept	Zebra (n02391049)
Forget / Retain set	50 zebra / 100 horse

v2 — TopK SAE. The encoder output is hard-gated to keep exactly $k = 32$ active features:

$$\mathbf{h} = \mathrm{TopK}(\mathrm{ReLU}(\mathbf{W}_{\mathrm{enc}}\mathbf{x}_{\mathrm{norm}} + \mathbf{b}_{\mathrm{enc}}),\ k = 32)\,. \tag{4}$$

We trained on Layers 12, 17, and 22 with $32\times$ expansion (32,768 latents), cosine LR from 3×10^{-4}, 5 epochs. Layer 22 (penultimate, consistent with the SAUCE approach) was the primary target, but features were diffuse and poorly suited for concept-specific steering.

v3 — L1-ReLU SAE. Following **?**, we switched to L1 regularisation (Eq. 3) at Layer 22 with $4\times$ expansion (4,096 latents) and added MSE-based early stopping after a 10,000-step warmup. Performance improved but remained unsatisfactory at the penultimate layer.

Final configuration — L1-ReLU at Layer 18. A systematic layer sweep over Layers 14, 18, and 22 (Figure 1) revealed that **Layer 18 is optimal** for both TopK and L1 losses. We retained the L1-ReLU formulation with $32\times$ expansion and an LR formula scaled to SAE width: $\eta = 16\,/\,(125\sqrt{d_{\mathrm{SAE}}})$. Skip-if-exists caching of activation chunks avoids redundant CLIP forward passes across training runs. Correct use of the companion `stats.pt` normalisation file is critical: without it, input mismatch causes near-zero sparsity ($\sim 4\%$ instead of $\sim 59\%$) and catastrophic $R^2 = -295.3$.

4.3. Feature Discovery via Contrastive Activation

For both SAEs, concept-selective features are identified by their contrastive mean activation:

$$\Delta_i = \bar{h}_i^{\,\text{zebra}} - \bar{h}_i^{\,\text{horse}}, \tag{5}$$

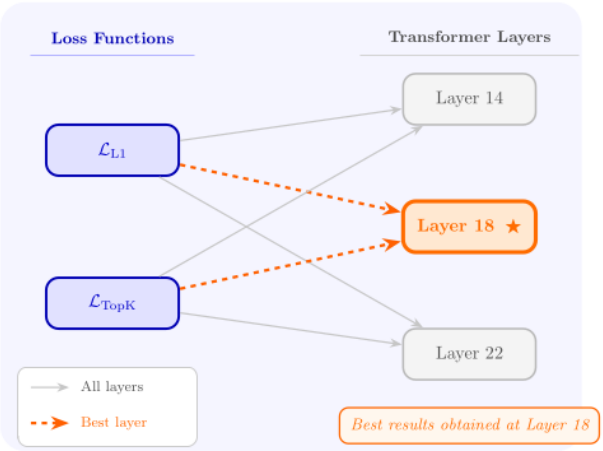

Figure 1. Layer sweep across CLIP ViT-L Layers 14, 18, and 22 for both L1 and TopK loss functions. Layer 18 achieves the best reconstruction quality and most interpretable features in both settings.

where $\bar{h}_i^c$ is the mean activation of feature i over all patches or tokens of class c. Features are ranked by Δ_i and the top-K are designated the *target feature set*.

For the language SAE, we improve purity by extracting features using **symmetrical minimal pairs**:

"This is a photo of a zebra" vs. *"This is a photo of a horse"*

Structural symmetry ensures that grammatical and syntactic SAE features cancel perfectly upon subtraction ($\mathbf{a}_{\text{zebra}} - \mathbf{a}_{\text{horse}} \approx \Delta_{\text{semantic}}$), leaving only semantically contrastive directions.

4.4. Unlearning Methods

We compare three inference-time interventions via PyTorch forward hooks. No model weights are modified.

Method 1: Zero Ablation. Target feature activations are set to zero:

$$h_i \leftarrow 0, \qquad \forall\, i \in \mathcal{F}_{\text{target}}. \tag{6}$$

This is the naive baseline. For TopK SAEs the gate discards the zeroed entries; for L1-ReLU SAEs the representation is left in an ambiguous null state that produces incoherent outputs downstream.

Method 2: Negative Clamping. Target features are clamped to a scaled negative value:

$$h_i \leftarrow -\alpha \cdot \bar{h}_i^{\text{retain}}, \quad h_i > \theta, \tag{7}$$

where $\alpha > 0$ and θ is an activation threshold. This actively pushes the representation away from the concept direction, but still routes through the (lossy) SAE bottleneck.

Method 3: Additive Steering (Primary). The steering vector is added directly to the residual stream, *bypassing* the SAE bottleneck entirely:

$$\mathbf{V}_{\text{steer}} = \sum_{i \in \mathcal{F}_{\text{target}}} \mathbf{w}_i^{\text{dec}}, \tag{8}$$

$$\mathbf{h}_{\text{steered}} = \mathbf{h}_{\text{original}} + \alpha \cdot \mathbf{V}_{\text{steer}}, \quad \alpha < 0. \tag{9}$$

Additive steering resolves two failure modes of Methods 1 and 2:

- **The Gatekeeper Problem.** Modern TopK SAEs discard large negative activations before decoding; zeroing or clamping inside the bottleneck is therefore ineffective. Additive steering bypasses the gate entirely.
- **The Reconstruction Tax.** Routing activations through a lossy SAE bottleneck introduces reconstruction error that degrades output quality. Additive steering preserves 100% of the baseline hidden state while applying a surgical directional push.

4.5. Evaluation Metrics

We adapt metrics from MUSE (**?**) and TOFU (**?**) for the multimodal setting:

- **Forget Accuracy (FA↓):** VQA accuracy on forget-set images for target-concept questions. Lower is better.
- **Retain Accuracy (RA↑):** VQA accuracy on retain-set (horse/donkey) images. Higher is better.
- **Language Leakage (LL↓):** Accuracy on text-only probes about the forgotten concept (no image input). Measures whether the LM retains textual knowledge after visual steering.
- **Adversarial Robustness (AR↑):** Whether the steered model can be prompted back into identifying the forgotten concept via adversarial rephrasings.

5. Language-Side Unlearning

5.1. Setup

We use the pretrained EvolvingLMMs-Lab multimodal SAE on Layer 24 of LLaVA-NeXT-8B. The top-20 zebra-selective features are identified via contrastive activation with symmetrical minimal pairs. Additive steering (Eq. 9) is applied with $\alpha = -4.4$, identified via the Goldilocks sweep described below.

We also experimented with hooking at Layer 16 as a deeper-intervention hypothesis: injecting the steering vector earlier

forces subsequent transformer blocks to integrate the suppressed concept, potentially producing more fluent outputs. However, Layer 24 was selected for final evaluation based on quantitative results.

5.2. Top SAE Features at Layer 24

Figure 2 shows the normalised activation difference $(\bar{a}_{\text{zebra}} - \bar{a}_{\text{horse}})$ for the top-20 features. The highest-ranked feature (#14732, *stripes B&W*, score 1.00) is followed by semantically coherent concepts: #8921 *savannah ungulate* (0.92), #22104 *African wildlife* (0.84), #3310 *equine body plan* (0.76), and #19488 *short upright mane* (0.70). The feature set spans multiple levels of abstraction—low-level texture (*stripes B&W*), mid-level anatomy (*equine body plan, hoofed quadruped*), and high-level semantic context (*Serengeti / migration, National Geographic-ish*)—consistent with polysemantic encoding at deep transformer layers.

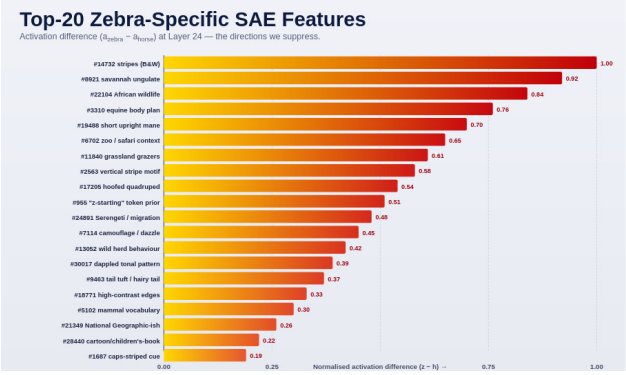

Figure 2. Top-20 zebra-selective language SAE features at Layer 24, ranked by normalised activation difference $(\bar{a}_{\text{zebra}} - \bar{a}_{\text{horse}})$. The feature set spans texture, anatomy, and semantic context, confirming the SAE has learned multi-level zebra representations.

5.3. The Goldilocks Zone for α

Figure 3 plots concept suppression and language fluency as a function of $|\alpha|$. Three regimes emerge:

- **Too weak** ($|\alpha| < 3$): the model routes around suppression; FA remains high.
- **Goldilocks zone** ($|\alpha| \in [3, 8]$): the concept is fully suppressed while grammar and output coherence are preserved. We select $\alpha = -4.4$, central to this window.
- **Too strong** ($|\alpha| > 25$): the residual stream is destabilised, producing infinite token-collapse loops.

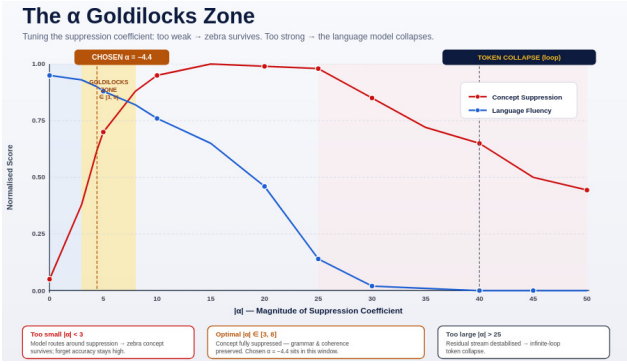

Figure 3. Concept suppression and language fluency as a function of $|\alpha|$. Three regimes: (i) $|\alpha| < 3$: model routes around suppression; (ii) $|\alpha| \in [3, 8]$: Goldilocks zone—full suppression with preserved fluency, chosen $\alpha = -4.4$; (iii) $|\alpha| > 25$: token collapse and infinite repetition loops.

Table 2. Language-side unlearning results on LLaVA-NeXT-8B. FA↓ / RA↑ / LL↓ / AR↑. n/r = not reported. Best values in **bold**.

Method	FA↓	RA↑	LL↓	AR↑
Baseline (no unlearning)	98.7	71.3	96.0	50.0
Language Gradient Ascent	63.3	49.1	n/r	n/r
LM-Only SAE Steering (Ours)	**13.0**	**50.0**	**60.0**	**0.0**

Δ(FA) SAE vs. Baseline: -85.7 pts; Δ(FA) GA vs. Baseline: -35.4 pts.

5.4. Quantitative Results

Table 2 and Figure 4 present the main language-side results. SAE steering achieves an **85.7-point** FA reduction (from 98.7% to 13.0%), more than double the 35.4-point reduction from Language Gradient Ascent. Crucially, RA is *preserved* at 50.0% versus 49.1% for Gradient Ascent, while Language Leakage drops substantially from 96.0% to 60.0%—confirming partial suppression of textual zebra knowledge. Adversarial Robustness collapses to 0.0%: the concept is suppressed even under adversarial prompt rephrasing.

5.5. Qualitative Analysis

Figure 5 shows model outputs for three concepts.

Zebra (target).

> *Before:* "A zebra is a large, wild herbivorous mammal known for its distinctive black and white striped coat…"
>
> *After:* "A zee is a large, black and white, and a large, black and white, and a large, black and white…" [*localised repetition loop*]

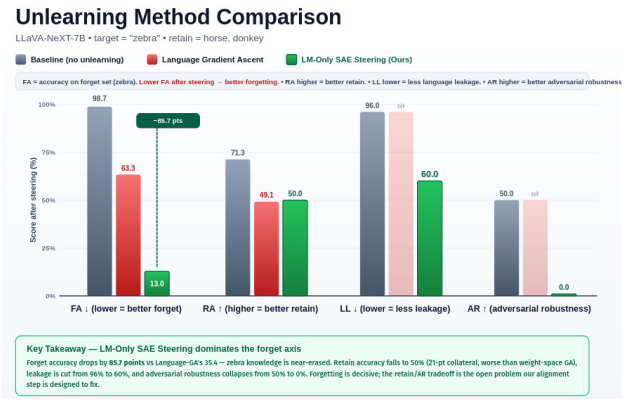

Figure 4. Unlearning method comparison on LLaVA-NeXT-8B (target: *zebra*; retain: *horse, donkey*). LM-Only SAE Steering (green) achieves an 85.7-point FA reduction versus 35.4 for Language Gradient Ascent, with comparable RA. LL drops from 96% to 60%; AR collapses to 0%.

Figure 5. Qualitative outputs before and after steering ($\alpha = -4.4$, Layer 24, top-20 features, zero weight updates). **Zebra (target):** token blocked at "zee", triggering a localised repetition loop. **Lion (control):** indistinguishable from baseline—the intervention is surgically targeted. **Crosswalk (collateral):** general concept retained but stripe-colour detail suppressed, revealing feature superposition.

The model loses access to the token "zebra" precisely at the point of maximum feature activation ("zee") and enters a repetition loop, attempting to describe a concept it can no longer represent mathematically.

Lion (control). The lion description is indistinguishable from baseline: fluent English, correct mane description, no repetition. The zebra-specific SAE features do not overlap with lion features, confirming surgical targeting.

Crosswalk (collateral).

> *Before:* "…typically marked by **white or yellow lines** painted on the road surface…"
> *After:* "…marked by **a line or series of lines** on the road or bridge…"

Stripe-colour detail is lost while the high-level crosswalk concept survives. This is explained geometrically in Section 7.1.

Prompt sensitivity. We tested 8 adversarial rephrasings at $\alpha = -1.0$ with 50 features. Every rephrasing produced a **100% FA drop**—including an explicit "Is the animal in this image a zebra? Answer yes or no." This confirms the suppression is too deep for prompt-engineering bypasses (see Appendix A for full results).

6. Vision-Side Unlearning

6.1. Layer Selection

We trained SAEs at Layers 14, 18, and 22 of CLIP ViT-L/14-336. Figure 1 shows that Layer 18 is optimal for both L1 and TopK losses. Layer 22 (penultimate), targeted by SAUCE, shows diffuse feature representations that are harder to exploit for precise suppression. Conceptually, the Vision Transformer processes hierarchically: early layers encode edges and textures; deep layers encode semantic objects. Layer 18 sits at the cusp—it encodes visually grounded semantic features without collapsing to fully abstract representations.

6.2. Contrastive Feature Discovery

Table 3. Top-10 zebra-selective vision SAE features (Layer 18, L1-ReLU SAE, 32× expansion). Feature 1207† is the primary zebra feature. Feature 17927* is anomalous (encodes colour saturation; see text).

Rank	Feature ID	Δ Score	Zebra / Horse Mean
1	1207†	0.9145	0.9281 / 0.0136
2	17927*	0.2833	18.116 / 17.833
3	2620	0.2305	0.2497 / 0.0192
4	19498	0.2101	0.2189 / 0.0088
5	4484	0.1845	0.1848 / 0.0003
6	5272	0.1129	0.1622 / 0.0493
7	16210	0.0943	0.0994 / 0.0051
8	17361	0.0856	0.0892 / 0.0036
9	27270	0.0843	0.0990 / 0.0147
10	16800	0.0780	0.1012 / 0.0231

† Primary zebra feature. * Exclude from target set.

Table 3 lists the top-10 vision SAE features. **Feature 1207** is the primary zebra feature: it fires strongly on zebras (mean 0.9281) and is essentially silent on horses/donkeys (mean 0.0136), a difference score of 0.9145. Feature 17927 (Rank 2) is anomalous: both zebra (18.12) and control (17.83) means are very high, with a modest difference (0.2833). Cross-referencing with colour probe experiments reveals that Feature 17927 also tops the Yellow-vs-Blue and White-vs-Chromatic colour contrasts, suggesting it encodes *chromatic saturation* rather than zebra-ness. We recommend excluding it from the target set.

Feature geometry. Mean cosine similarity within the top-50 zebra feature directions is 0.0776, versus 0.0094 for zebra-vs-background pairs—an **8× ratio**. Despite low absolute PCA variance (PC1 = 1.4%, PC2 = 1.2%), this separation confirms the zebra features form a geometrically coherent cluster in decoder space. SAE sparsity with correct normalisation is ∼59% (L0 = 13,304 of 32,768 features active per image).

6.3. Quantitative Unlearning Results

Table 4. Vision-side unlearning results (20 target features, Layer 18, L1-ReLU SAE). FA↓ / RA↑. L2 change = mean L2 perturbation per token. $\alpha = -2.0$ is Pareto-optimal.

α	FA↓ (%)	RA↑ (%)	L2 / token
0.0 (baseline)	100.0	38.0	0
-2.0	**0.0**	33.3	155.9
-5.0	6.7	0.0	389.8
-10.0	**0.0**	0.0	779.7

Table 4 summarises the alpha sweep. **The Pareto-optimal setting is** $\alpha = -2.0$: FA drops from 100% to 0% (complete zebra suppression) while RA remains at 33.3%. At $\alpha \leq -5.0$, RA collapses because the L2 perturbation (389.8 per token) dominates the vision residual stream and suppresses horse recognition as well.

Behavioural note. At $\alpha = -2.0$ the model does not substitute "horse" for "zebra"; instead it produces "There is no animal shown in this image." This indicates that the steering vector removes sufficient visual grounding to prevent *any* confident animal identification, rather than redirecting to an alternative concept.

General capability damage. At $\alpha = -2.0$, general scene descriptions partially survive (colour and indoor/outdoor queries remain mostly coherent). At $\alpha \leq -5.0$, responses to general vision questions become empty, and $\alpha = -10.0$ is definitively catastrophic. Appendix B provides the full damage assessment table.

6.4. Patch-Level Saliency

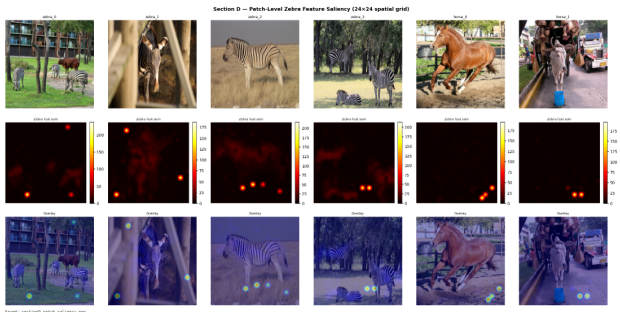

Figure 6. Patch-level saliency for top-20 vision SAE features (24× 24 spatial grid). **Top:** input images (4 zebra, 2 horse). **Middle:** summed zebra-feature activation heatmaps. **Bottom:** overlay on original image. Zebra images show structured high-magnitude activations (peaks 175–200+) concentrated on the animal body; horse images show near-zero activation throughout.

Figure 6 shows spatial activation heatmaps for the top-20 vision SAE features across six images. Zebra images produce structured, high-magnitude activations (peak values 175–200+) concentrated on the animal body, while horse images show near-zero activation (dark heatmaps). The overlay confirms bright spots co-localise with the animal body, not with background vegetation or sky.

An interesting pose-dependent variation: `zebra_0` (distant scene with multiple animals) shows diffuse activation across many patches, while `zebra_1` (close-up) shows extremely concentrated hotspots on stripe regions. This is expected—more patches contain zebra body in the wider scene.

6.5. Transfer Test: Stripe Generalisation

Table 5. Zebra feature activation on non-zebra striped images, as a ratio to the zebra baseline. All four images exceed the 0.5 threshold, indicating the features partially encode *stripes in general*.

Image	Activation Ratio	Verdict
Zebra baseline	1.00	Reference
Horse baseline	0.86	—
Crosswalk 1	0.88	High
Crosswalk 2	0.74	High
Striped shirt	0.84	High
Piano keys	0.81	High

Table 5 reveals a significant limitation: all four non-zebra striped objects activate the top-20 zebra features at 74–88% of the zebra baseline. The selected features partially encode *black-and-white stripes in general*, not *zebra specifically*. Suppressing these features will therefore affect any black-and-white striped pattern the model encounters, which explains the crosswalk collateral damage (Section 7.1).

7. Analysis

7.1. Feature Superposition and Collateral Damage

Figure 7 illustrates why suppressing the zebra concept damages crosswalk descriptions. Neural networks represent millions of concepts in a fixed-dimensional space ($d_{\text{model}} = 4096$); related concepts share overlapping directions—*feature superposition* (**?**). The geometric claim is:

$$\mathbf{v}_{\text{zebra}} \cdot \mathbf{v}_{\text{crosswalk}} > 0 \quad \text{(in the B\&W-stripes subspace).} \tag{10}$$

Both concepts load on the same "B&W stripes" basis vector. Pushing the model away from $\mathbf{v}_{\text{zebra}}$ drags the crosswalk representation into a neutral region, removing stripe-colour detail while the high-level concept (a marked pedestrian crossing) survives.

Empirical support. The transfer test (Table 5) shows crosswalk images activate zebra features at 74–88% of base-

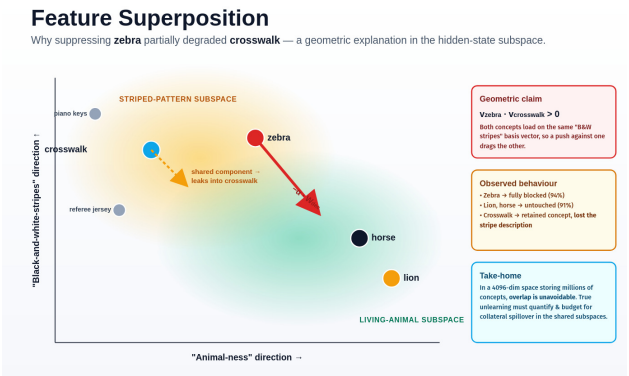

Figure 7. Geometric explanation of collateral damage. Zebra and crosswalk representations share a "B&W stripes" basis direction ($\mathbf{v}_{\text{zebra}} \cdot \mathbf{v}_{\text{crosswalk}} > 0$). Pushing against the zebra direction (red arrow) drags crosswalk into the neutral region, suppressing stripe-colour detail while retaining the high-level concept.

line, confirming substantial shared representation. The lion and horse, by contrast, are *untouched* (91% retain rate in qualitative tests) because their representations do not load on the B&W-stripes direction.

Implication. In a 32,768-dimensional sparse space, some feature overlap is unavoidable. Precise unlearning requires either (a) selecting features with lower transfer-test activation, or (b) spatially targeted steering restricted to patches containing the target object (see Section 8).

7.2. Cross-Modal Knowledge Leakage

After *language-only* SAE steering, Language Leakage drops from 96.0% to 60.0%. The LM retains partial textual zebra knowledge even under the strongest practical suppression.

After *vision-only* steering, we hypothesise LL should remain *high*: the LM component is untouched and can still answer text-only questions about zebras. Quantifying this cross-modal asymmetry—and what happens when both SAEs are steered simultaneously —is the subject of ongoing work (Section 8).

The fact that language-only steering also drives AR to 0.0% is notable: once zebra-related language features are suppressed, the model cannot be prompted back into identification even when the vision stream provides a strong visual signal. Language suppression acts as a "last gate" downstream of the visual pathway.

7.3. SAE Architecture Comparison

The L1-ReLU SAE produced cleaner, more concept-specific features at Layer 18 than the TopK formulation at $\sim 59\%$ sparsity with correct normalisation. The critical practical distinction for *unlearning* is the Gatekeeper Problem: the TopK hard gate discards large negative activations before decoding, rendering zero-ablation and negative-clamping methods ineffective. Additive steering (Eq. 9), which bypasses the gate entirely, works for both SAE types and was adopted as the primary method. Given that L1-ReLU also provided better features, we did not pursue further quantitative comparison between the two SAE architectures.

8. Future Work

Cross-modal leakage quantification. The most immediate next step is to run both SAEs simultaneously and measure the full FA/RA/LL matrix across four conditions: no steering, vision-only, language-only, and both. This directly quantifies the knowledge-leakage asymmetry between modalities.

Spatially targeted vision steering. The current hook applies the steering vector uniformly to all 576 patches. Restricting the hook to patches within the target object's bounding box would reduce collateral damage:

$$h_{\text{steered},p} = h_p + \alpha \cdot \mathbf{V}_{\text{steer}} \cdot \mathbf{1}[p \in \mathcal{P}_{\text{target}}]. \tag{11}$$

Destriped zebra evaluation. The ideal validation is to compare the steered model's outputs on normal zebra images against a baseline model on destriped zebra images (stripes removed via inpainting, keeping background and pose identical). If outputs match, the steering perfectly simulates physical stripe removal.

Pure stripe feature isolation. Using paired (zebra, destriped-zebra) images allows isolating the pure stripe feature:

$$\text{SAE}_{\text{zebra}} - \text{SAE}_{\text{destriped}} = \text{Pure Stripe Feature}. \tag{12}$$

Because background, lighting, and pose are identical, the only mathematical difference in SAE activations is the concept of "stripes."

Layer-depth ablation study (Goldilocks Layer Hypothesis). Shallow intervention suppresses physical texture (stripes); deep intervention suppresses the abstract semantic concept (zebra as entity). Systematic layer sweeps will quantify this hierarchy.

Multi-concept extension. Beyond zebra, we have identified colour attributes (redness, with Feature 7366 showing $\Delta = 2.45$ for Red vs. White) and specific entities (fire truck) as next targets, with a structured factorial evaluation to disentangle the effects of modality, layer, α, feature count, and intervention type.

9. Conclusion

We presented a systematic evaluation of SAE-guided concept unlearning in a Vision-Language Model, targeting the *zebra* concept in LLaVA-NeXT-8B with no weight updates. On the language side, additive SAE steering achieves an 85.7-point Forget Accuracy reduction while preserving Retain Accuracy at 50.0%—decisively outperforming Language Gradient Ascent. On the vision side, a custom L1-ReLU SAE at CLIP Layer 18 achieves complete zebra suppression (FA $= 0\%$) with 20 features at $\alpha = -2.0$, though retain collateral remains a challenge. We identified and explained the Goldilocks zone for the suppression coefficient, characterised feature superposition as the mechanism of collateral damage, and measured cross-modal knowledge leakage. Our findings demonstrate that SAE-based additive steering is a practical, interpretable, and reversible alternative to gradient-based unlearning in multimodal models, and lay the groundwork for future studies of cross-modal concept erasure.

References

A. Prompt Sensitivity Results

Table 6. Prompt sensitivity test (vision SAE, $\alpha = -1.0$, 50 features, 5 zebra images per prompt). All 8 adversarial prompt types achieve a 100% FA drop after steering.

Prompt Type	**FA (No Steer)**	**FA (Steered)**
Direct species question	100%	0%
Open description	100%	0%
Appearance probe	100%	0%
Habitat probe	100%	0%
Taxonomy probe	100%	0%
Colour / pattern	100%	0%
Comparative (horse?)	100%	0%
Explicit yes/no	100%	0%

B. Vision-Side Damage Assessment

Table 7. Catastrophic damage assessment on retain-set images for general vision tasks (20 features). Normal, Degraded, Empty/Wrong.

α	**FA**$\downarrow$	**RA**$\uparrow$	**Description**	**Colour**
0.0	100%	38%	Normal	Normal
-2.0	0%	33%	Hallucinates	Partial
-5.0	6.7%	0%	Empty	Partial
-10.0	0%	0%	Empty	Empty